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Connector device

Abstract

A connector device has a constitution in a connector in which a plurality of terminals protrude into a connector opening. The plurality of terminals are respectively provided with a plurality of switches that have independent states of supplying power to the terminals and are switchable in between. A plurality of conductor plates are provided in the connector opening in an attachable and detachable state, and are each configured to electrically connect at least two of the plurality of terminals. The plurality of conductor plates are coupled by an insulating member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-144778 filed on Sep. 12, 2022, the contents of which are incorporated herein by reference.
TECHNICAL FIELD
(2) The present disclosure relates to a connector device.
(3) Various types of electronic components are disposed at various portions of a vehicle. Electronic control units (ECUs) mounted on portions of a vehicle are required to control various electronic components to be controlled, that is, on and off of loads, magnitude of output and the like, with switch circuits.
(4) For example, in a configuration shown in FIG. 2 of JP2012-236472A, an output circuit including four semiconductor switch elements Q**1**H, Q**1**L, Q**2**H, and Q**2**L is used to bidirectionally control a current flowing through an electric motor **20** that is a load. An ECU including such an output circuit (switch circuit) is generally connected to a load via a wire harness including a predetermined connector.
(5) However, switch circuits provided in ECUs are required to comply with types and specifications of loads. For example, when controlling an electric motor requiring bidirectional

current control, four switch elements are necessary for one load, as in the output circuit 40 in JP2012-236472A. On the other hand, when controlling loads such as an electric motor, a lamp, and a heater that do not require switching of a driving direction, each of the loads can be controlled by one switch alone.

(6) In addition, a switch for applying a high voltage to a positive terminal of a load may be necessary, or a switch for applying a low voltage to a negative terminal of a load may be necessary, depending on a type and a specification of a load. Accordingly, circuit configurations of ECUs are designed to change in various ways in accordance with differences in location where the ECUs are disposed, differences in specification between vehicles and the like.

(7) On the other hand, it is desirable that the configurations of the ECUs be commonized as much as possible regardless of differences in type, grade and the like between vehicles. Such commonization of the circuit configurations contributes to reduction in ECU part numbers, reduction in manufacturing costs of ECUs, commonization of attachment work and the like.

(8) When an ECU that can be commonly used in vehicles of various specifications is designed, however, a switch circuit that can cope with specifications of all types of loads is required. Accordingly, the number of components to be installed increases compared to a minimum number of components required in design.

(9) For example, four switch elements are required to be installed for one load as in the output circuit 40 in JP2012-236472A. On the other hand, only one or two of the four switch elements may be used, depending on a type of a load actually connected to the output of an ECU. In this case, the remaining two or three switching elements of the four switching elements are uselessly attached components that are not used at all. Since such uselessly attached components are wasted, an increase in the number of components leads to an increase in the cost of the ECU.

(10) The present disclosure is made in view of the above circumstance, and an object of the present disclosure is to provide a connector device that can reduce waste components that are uselessly attached when circuit configurations are commonized in electronic control units or the like in a vehicle.

SUMMARY OF INVENTION

(11) A connector device has a constitution in a connector in which a plurality of terminals protrude into a connector opening. The plurality of terminals are respectively provided with a plurality of switches that have independent states of supplying power to the terminals and are switchable in between. A plurality of conductor plates are provided in the connector opening in an attachable and detachable state, and are each configured to electrically connect at least two of the plurality of terminals. The plurality of conductor plates are coupled by an insulating member.

(12) The present disclosure is briefly described above. Details of the present disclosure can be further clarified by reading modes for carrying out the disclosure (hereinafter referred to as “embodiments”) described below with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a perspective view showing an appearance of a main portion of a connector device according to an embodiment of the present disclosure;

(2) FIG. **2** is a perspective view showing inside of a connector to which a joint plate is installed;

(3) FIG. **3** is an exploded perspective view showing the joint plate;

(4) FIG. **4** is an electric circuit diagram showing a configuration example of an electronic control unit using the connector device of the present disclosure;

(5) FIG. **5** is an electric circuit diagram showing a configuration example of an electronic control unit using the connector device of the present disclosure;

(6) FIG. **6** is an electric circuit diagram showing a configuration example of an electronic control unit not using the connector device of the present disclosure; and

(7) FIG. **7** is an electric circuit diagram showing a configuration example of an electronic control unit not using the connector device of the present disclosure.

DESCRIPTION OF EMBODIMENTS

(8) A specific embodiment of the present disclosure will be described below with reference to the drawings.

(9) FIG. **1** is a perspective view showing an appearance of a main portion of a connector device according to the embodiment of the present disclosure. FIG. **2** is a perspective view showing inside of a connector to which a joint plate **20** is installed. FIG. **3** is an exploded perspective view showing the joint plate **20**.

(10) As shown in FIG. **1**, a connector **10** includes a connector housing **11** and a plurality of metal terminals **13***a*, **13***b*, **13***c*, and **13***d* accommodated in the connector housing **11**. The connector **10** is male and has a rectangular connector opening **12**.

(11) The metal terminals **13***a* to **13***d* are arranged in parallel to each other, and protrude in an elongated pin shape into the connector opening **12** in a direction of an arrow A**1** shown in FIG. **1**. In the present embodiment, each of the metal terminals **13***a* to **13***d* has a rectangular cross-sectional shape. Lower end sides of the metal terminals **13***a* to **13***d* are connected to a circuit such as a printed circuit board.

(12) The connector **10** according to the present embodiment further includes the joint plate shown in FIG. **1**. The joint plate **20** has a thin flat plate shape, and has an outer shape that is substantially the same as a cross-sectional shape of an inner peripheral wall of the connector opening **12**. The joint plate **20** is attachable to and detachable from the connector opening **12** of the connector housing **11**. Plural joint plates **20** of different types may be prepared to be selected appropriately according to the use of the connector **10**.

(13) As shown in FIG. **3**, the joint plate **20** according to the present embodiment includes two copper plates **21**, **22** and an insulating member **23**. Each of the copper plates **21**, **22** has a thin plate shape, and has a direction of an arrow A**2** as a longitudinal direction.

(14) The copper plate **21** has two through holes **21***a*, **21***b* through which the metal terminals **13***a*, **13***b* can be inserted, respectively. The through holes **21***a*, **21***b* have shapes and sizes that are the same as outer shapes of the metal terminals **13***a*, **13***b*. Similarly, the copper plate **22** has two through holes **22***a*, **22***b* through which the metal terminals **13***c*, **13***d* can be inserted, respectively. The through holes **22***a*, **22***b* have shapes and sizes that are the same as outer shapes of the metal terminals **13***c*, **13***d*.

(15) As shown in FIG. **3**, in a state in which the copper plate **21**, the insulating member **23**, and the copper plate **22** are arranged in a direction of an arrow A**3**, these components can be coupled laterally to be integrated. In the example shown in FIG. **3**, the copper plate **21** is formed with concave portions **21***c*, **21***d* in a side surface thereof. The copper plate **22** is formed with a convex portion **22***c* and a convex portion **22***d* on a side surface thereof. The insulating member **23** is formed with convex portions **23***a*, **23***b* on a front side surface thereof, and is formed with concave portions **23***c*, **23***d* in a back side surface thereof.

(16) The copper plate **21** and the insulating member **23** are coupled by aligning and fitting the concave portion **21***c* and the convex portion **23***a* as well as the concave portion **21***d* and the convex portion **23***b*, respectively. The copper plate **22** and the insulating member **23** are coupled by fitting the concave portion **23***c* and the convex portion **22***c* as well as the concave portion **23***d* and the convex portion **22***d*, respectively. The joint plate **20**, in which the copper plates **21**, **22** and the insulating member **23** are coupled and integrated, is inserted and installed into the connector opening **12** of the connector housing **11** as shown in FIG. **1**. In the embodiment described in FIG. **3**, the copper plate **21** has the concave portions **21***c*, **21***d*. The copper plate **21** may have convex portions that are fitted to concave portions of the insulating member **23**. Similarly, the copper plate

22 may have concave portions that are fitted to convex portions of the insulating member **23**.
(17) When the joint plate **20** is installed to the connector housing **11**, the metal terminals **13***a* to **13***d* and the joint plate **20** are coupled inside the connector housing **11** as shown in FIG. **2**. That is, the metal terminals **13***a* to **13***d* penetrate the through holes **21***a*, **21***b*, **22***a*, and **22***b*, respectively.
(18) The copper plates **21**, **22** and the metal terminals **13***a* to **13***d* are formed of a conductor that has high conductivity, and thus the copper plate **21** and the metal terminal **13***a* are electrically connected, and the copper plate **21** and the metal terminal **13***b* are electrically connected. Further, the copper plate **22** and the metal terminal **13***c* are electrically connected, and the copper plate **22** and the metal terminal **13***d* are electrically connected.
(19) Accordingly, electric circuits of the two metal terminals **13***a*, **13***b* are electrically connected (short-circuited) through the copper plate **21**. Similarly, electric circuits of the two metal terminals **13***c*, **13***d* are electrically connected through the other copper plate **22**. Since the insulating member **23** is interposed between the two copper plates **21**, **22**, the electric circuits of the metal terminals **13***a*, **13***b* are electrically insulated from the electric circuits of the metal terminals **13***c*, **13***d*.
(20) The copper plates **21**, **22** are provided with, at portions **20***a* in the vicinity of peripheral walls of the through holes **21***a*, **21***b*, **22***a*, and **22***b*, elastic members such as metal springs to maintain a good contact state of contact points with the metal terminals **13***a* to **13***d*. The contact state between the copper plates **21**, **22** and the metal terminals **13***a* to **13***d* may also be maintained by a press-fitting structure or the like.
(21) Even when the joint plate **20** is installed to the connector opening **12** of the connector housing **11**, top ends of the metal terminals **13***a* to **13***d* protrude further upward of the joint plate **20**. Accordingly, a counterpart female connector (not shown) can be fitted and coupled to the connector opening **12** of the connector **10** after the joint plate **20** is installed to the connector housing **11**. Electronic components of various loads can be connected to the female connector through a wire harness.
(22) <Configuration of Electronic Control Unit Using Connector Device>
(23) FIGS. **4** and **5** show electric circuit configurations of two types of electronic control units **100**A, **100**B using the connector **10** shown in FIGS. **1** to **3**. The electronic control unit **100**A shown in FIG. **4** is designed on an assumption of controlling energization of an electric motor that requires bidirectional drive control as a load L**1**.
(24) The electronic control unit **100**A includes four independent semiconductor switch elements Q**1**, Q**2**, Q**3**, and Q**4**, and a control unit **35**. The two semiconductor switch elements Q**1**, Q**3** constitute Hi side output circuits **31**, **33**, respectively, and the remaining two semiconductor switch elements Q**2**, Q**4** constitute Lo side output circuits **32**, **34**, respectively.
(25) The Hi side output circuit **31** can supply a high voltage to an output side when the semiconductor switch element Q**1** is conducted. The Lo side output circuit **32** can supply a low voltage to an output side when the semiconductor switch element Q**2** is conducted. The Hi side output circuit **33** can supply a high voltage to an output side when the semiconductor switch element Q**3** is conducted. The Lo side output circuit **34** can supply a low voltage to an output side when the semiconductor switch element Q**4** is conducted.
(26) In the electronic control unit **100**A, outputs of the Hi side output circuits **31**, **33** and the Lo side output circuits **32**, **34** are connected to the connector **10**. The output of the Hi side output circuit **31** is connected to the metal terminal **13***a* of the connector **10**, the output of the Lo side output circuit **32** is connected to the metal terminal **13***b* of the connector **10**, the output of the Hi side output circuit **33** is connected to the metal terminal **13***c* of the connector **10**, and the output of the Lo side output circuit **34** is connected to the metal terminal **13***d* of the connector **10**.
(27) The joint plate **20** is installed to the connector **10**. Accordingly, the metal terminals **13***a*, **13***b* of the connector **10** are electrically connected by the copper plate **21**, and the metal terminals **13***c*, **13***d* are electrically connected by the copper plate **22**.
(28) The load L**1** is connected to the connector **10** through a predetermined wire harness. The load

L**1** has a positive terminal and a negative terminal connected to the metal terminals **13***a*, **13***d* of the connector **10**, respectively. The positive terminal of the load L**1** is also connected to the metal terminal **13***b* through the copper plate **21**, and the negative terminal of the load L**1** is also connected to the metal terminal **13***c* through the copper plate **22**.
(29) When the control unit **35** controls the semiconductor switch elements Q**1**, Q**2**, Q**3**, and Q**4** to be in a conducted, non-conducted, non-conducted, and conducted state, respectively, a current flows through the load L**1** in a forward direction. When the control unit **35** controls the semiconductor switch elements Q**1**, Q**2**, Q**3**, and Q**4** to be in a non-conducted, conducted, conducted, and non-conducted state, a current flows through the load L**1** in a reverse direction.
(30) On the other hand, the electronic control unit **100**B shown in FIG. **5** is designed on an assumption of separately controlling energization of four loads L**2**, L**3**, L**4**, and L**5**. In the example of FIG. **5**, the two loads L**2**, L**4** are electric motors that require application of a high voltage. The two loads L**3**, L**5** are lamps (LP) that require application of a low voltage.
(31) Similarly to the electronic control unit **100**A of FIG. **4**, an internal circuit of the electronic control unit **100**B includes the Hi side output circuit **31**, the Lo side output circuit **32**, the Hi side output circuit **33**, the Lo side output circuit **34**, and the control unit **35**. That is, the two types of electronic control units **100**A, **100**B use a circuit having a common configuration.
(32) However, in the electronic control unit **100**B of FIG. **5**, the joint plate **20** is not installed to the connector **10**. Accordingly, the electric circuits of the four metal terminals **13***a* to **13***d* of the connector **10** are each in an independent state.
(33) In the electronic control unit **100**B shown in FIG. **5**, the control unit **35** can apply a high voltage to a terminal of the load L**2** and drive the load L**2** by controlling the semiconductor switch element Q**1** to a conducted state. The control unit **35** can apply a low voltage to a terminal of the load L**3** and drive the load L**3** by controlling the semiconductor switch element Q**2** to a conducted state.
(34) The control unit **35** can apply a high voltage to a terminal of the load L**4** and drive the load L**4** by controlling the semiconductor switching element Q**3** to a conducted state. The control unit **35** can apply a low voltage to a terminal of the load L**5** and drive the load L**5** by controlling the semiconductor switch element Q**4** to a conducted state.
(35) That is, by connecting the electric circuits as in the electronic control unit **100**A of FIG. **4** and the electronic control unit **100**B of FIG. **5** using the connector **10** as shown in FIGS. **1** to **3**, the loads L**1** to L**5** having different specifications can be appropriately controlled even when electric circuits having the same configuration in the units are used. Since all of the four semiconductor switch elements Q**1** to Q**4** installed in the units can be effectively used, waste components that are uselessly attached can be reduced.
(36) <Configuration of Electronic Control Unit Using General Connector>
(37) FIGS. **6** and **7** show electric circuit configurations of two types of electronic control units **100**C, **100**D using a general connector **50**.
(38) In the electronic control unit **100**C shown in FIG. **6**, the load L**1** is connected to an output side of the connector **50** having a general structure. Similarly, to the electronic control unit **100**A of FIG. **4**, a current is required to flow through the load L**1** in two directions. For this reason, in an internal circuit of the electronic control unit **100**C, outputs of the two semiconductor switch elements Q**1**, Q**2** are electrically connected in common upstream of the connector **50**. Outputs of the two semiconductor switch elements Q**3**, Q**4** are electrically connected in common upstream of the connector **50**.
(39) On the other hand, the electronic control unit **100**D shown in FIG. **7** is designed on a premise that an internal circuit having the same configuration as that of the electronic control unit **100**C of FIG. **6** is used. In the electronic control unit **100**D of FIG. **7**, two types of loads L**2**, L**3** are connected to the output side of the connector **50**.
(40) Here, the load L**2** is required to be driven by application of a high voltage, and thus the

electronic control unit **100**D connects the load L**2** to an output side of the semiconductor switch element Q**1** through the connector **50**. The load L**3** is required to be driven by application of a low voltage, and thus the electronic control unit **100**D connects the load L**3** to an output side of the semiconductor switch element Q**4** through the connector **50**.
(41) In the configurations of FIGS. **6** and **7**, the electronic control units **100**C, **100**D can use a common internal circuit. However, outputs of the semiconductor switch elements Q**1**, Q**2** are connected to the common circuit in advance and outputs of the semiconductor switch elements Q**3** and Q**4** are connected to the common circuit in advance, and thus this configuration cannot be changed. For this reason, although the electronic control unit **100**D of FIG. **7** uses the semiconductor switch element Q**1** for driving the load L**2** and uses the semiconductor switch element Q**4** for driving the load L**3**, the remaining two semiconductor switch elements Q**2**, Q**3** are not effectively used. That is, the two semiconductor switch elements Q**2**, Q**3** and components related thereto are uselessly attached, and the cost of the device is expected to increase by the amount of waste components installed in the electronic control unit **100**D.
(42) As described above, the connector **10** according to the present embodiment can use a circuit unit having a common internal circuit configuration when connecting loads of different types and specifications to an output side as in the electronic control units **100**A, **100**B of FIGS. **4** and **5**. In addition, the circuit configuration can be changed as necessary at the connector by attaching, detaching or replacing the joint plate **20** installed to the connector **10**. For this reason, all of the plural semiconductor switch elements Q**1** to Q**4** installed in the internal circuit as in the electronic control units **100**A, **100**B can be effectively used. That is, the number of components that are uselessly attached such as the semiconductor switching elements Q**2**, Q**3** in FIG. **7** can be reduced. Accordingly, even when specifications of a vehicle or specifications of equipment to be mounted are frequently changed as in a case of an in-vehicle device, an ECU having a common configuration can be used without design changes, and the number of components that are uselessly attached can be reduced.
(43) The present disclosure is not limited to the above-described embodiment, and can be appropriately modified, improved and the like. In addition, materials, shapes, sizes, numbers, arrangement positions and the like of components in the above-described embodiment are freely selected and are not limited as long as the present disclosure can be implemented.
(44) For example, in the joint plate **20** installed to the connector **10** shown in FIGS. **1** to **3**, the two metal terminals **13***a*, **13***b* are commonly connected by the copper plate **21**, and the two metal terminals **13***c*, **13***d* are commonly connected by the copper plate **22** at the same time. Alternatively, for example, the copper plate **22** may be replaced with an insulating member, the two metal terminals **13***a*, **13***b* may be connected by the copper plate **21**, and the circuits of the two metal terminals **13***c*, **13***d* may be used independently. Alternatively, three metal terminals may be commonly connected by the L-shaped copper plate **21**, and the remaining one metal terminal alone may penetrate the insulating member. Plural types of joint plates **20** having different configurations may be prepared in advance and be switched between the types and used as necessary.
(45) Here, features of the connector device according to the embodiment of the present disclosure described above are briefly summarized and listed below.
(46) According to an aspect of the present disclosure, a connector device has a constitution (joint plate **20**) in a connector (**10**) in which a plurality of terminals (metal terminals **13***a* to **13***d*) protrude into a connector opening (**12**). The plurality of terminals are respectively provided with a plurality of switches (semiconductor switch elements Q**1** to Q**4**) that have independent states of supplying power to the terminals and are switchable in between. A plurality of conductor plates (copper plates **21**, **22**) are provided in the connector opening in an attachable and detachable state, and are each configured to electrically connect at least two of the plurality of terminals. The plurality of conductor plates (copper plates **21**, **22**) are coupled by an insulating member (**23**).
(47) According to the connector device having the above configuration, presence and absence of

the electrical connection between the plurality of terminals can be switched by attaching and detaching the conductor plates to and from the connector opening. Accordingly, it is possible to perform control corresponding to differences in specification between loads connected downstream of the connector while maintaining a common configuration of an upstream electric circuit (ECU or the like) including the plurality of switches. For this reason, components (switches or the like) installed in the ECU having a common configuration can be easily prevented from being uselessly attached.
(48) According to an aspect of the present aspect, an area of the plurality of conductor plates and the insulating member is substantially the same size as an area of the connector opening (**12**) (see FIG. **1**), when viewed in an extending direction (direction A**1**) of the terminals in a state in which the plurality of conductor plates are coupled by the insulating member.
(49) According to the connector device having the above configuration, the plurality of conductor plates and the insulating member coupled to each other can be easily and firmly fixed to the connector opening by commonizing the areas.
(50) According to an aspect of the present disclosure, the plurality of switches include a first switch (semiconductor switch element Q**1**) to output a high-level voltage to a first terminal (metal terminal **13***a*), and a second switch (semiconductor switch element Q**2**) to output a low-level voltage to a second terminal (metal terminal **13***b*). The conductor plates constitute a joint member (joint plate **20**) that electrically connects the first terminal and the second terminal in a state of being installed in the connector opening.
(51) According to the connector device having the above configuration, a load requiring the high-level voltage can be driven using the output of the first switch, and a load requiring the low-level voltage can be driven using the output of the second switch. Further, by installing the joint member to the connector opening, the first switch and the second switch can be combined, and a current can flow bidirectionally through a load requiring switching of a driving direction.
(52) According to an aspect of the present disclosure, the plurality of switches include a first switch (semiconductor switch element Q**1**) to output a high-level voltage to a first terminal (metal terminal **13***a*), a second switch (semiconductor switch element Q**2**) to output a low-level voltage to a second terminal (metal terminal **13***b*), a third switch (semiconductor switch element Q**3**) to output a high-level voltage to a third terminal (metal terminal **13***c*), and a fourth switch (semiconductor switch element Q**4**) to output a low-level voltage to a fourth terminal (metal terminal **13***d*). The conductor plates include, in a state of being installed to the connector opening, a first joint member (copper plate **21**) that electrically connects the first terminal and the second terminal, a second joint member (copper plate **22**) that electrically connects the third terminal and the fourth terminal, and the insulating member (**23**) that physically couples the first joint member and the second joint member in a state in which the first joint member and the second joint member are electrically insulated.
(53) According to the connector device having the above configuration, an H-bridge circuit as shown in FIG. **4** can be implemented by a combination of the first switch, the second switch, the third switch, and the fourth switch. This facilitates control of a load such as an electric motor that requires switching of a driving direction.
(54) According to an aspect of the present disclosure, the plurality of switches include a first switch to output a high-level voltage to a first terminal, a second switch to output a low-level voltage to a second terminal, a third switch to output a high-level voltage to a third terminal, and a fourth switch to output a low-level voltage to a fourth terminal. The conductor plates include, in a state of being installed to the connector opening, a first joint member that electrically connects the first terminal and the second terminal, a second joint member that electrically connects the third terminal and the fourth terminal, and the insulating member that physically couples the first joint member and the second joint member in a state in which the first joint member and the second joint member are electrically insulated. The first joint member (copper plate **21**) has a first through hole

(through hole **21***a*) and a second through hole (through hole **21***b*) through which the first terminal and the second terminal are inserted, respectively. The second joint member (copper plate **22**) has a third through hole (through hole **22***a*) and a fourth through hole (through hole **22***b*) through which the third terminal and the fourth terminal are inserted, respectively. The first joint member has a first concave portion or a first convex portion (concave portion **21***c*, concave portion **21***d*) that are fitted to the insulating member in a direction intersecting an insertion direction of the terminals. The second joint member has a second concave portion or a second convex portion (convex portion **22***c*, convex portion **22***d*) that are fitted to the insulating member in the direction intersecting with the insertion direction of the terminals.
(55) According to the connector device having the above configuration, it is possible to implement a conductor plate having an appropriate configuration necessary for constituting an H-bridge circuit with a combination of the first switch, the second switch, the third switch, and the fourth switch. The first joint member and the insulating member can be coupled, and the second joint member and the insulating member can be coupled.

Claims

1. A connector device having a constitution in a connector in which a plurality of terminals protrude into a connector opening, wherein the plurality of terminals are respectively provided with a plurality of switches for independently switching states of power supply to the terminals, a plurality of conductor plates are provided in the connector opening in an attachable and detachable state, and are each configured to electrically connect at least two of the plurality of terminals together the plurality of conductor plates are coupled with each other by an insulating member, and an area of the plurality of conductor plates and the insulating member is substantially the same size as an area of the connector opening, when viewed in an extending direction of the terminals in a state in which the plurality of conductor plates are coupled by the insulating member.
2. The connector device according to claim 1, wherein the plurality of switches include a first switch to output a high-level voltage to a first terminal, and a second switch to output a low-level voltage to a second terminal, and one of the conductor plates constitutes a joint member that electrically connects the first terminal and the second terminal in a state of being installed in the connector opening.
3. The connector device according to claim 1, wherein the plurality of switches include a first switch to output a high-level voltage to a first terminal, a second switch to output a low-level voltage to a second terminal, a third switch to output a high-level voltage to a third terminal, and a fourth switch to output a low-level voltage to a fourth terminal, and the conductor plates include, in a state of being installed in the connector opening, a first joint member that electrically connects the first terminal and the second terminal, a second joint member that electrically connects the third terminal and the fourth terminal, and the insulating member that physically couples the first joint member and the second joint member in a state in which the first joint member and the second joint member are electrically insulated.
4. The connector device according to claim 1, wherein the plurality of switches include a first switch to output a high-level voltage to a first terminal, a second switch to output a low-level voltage to a second terminal, a third switch to output a high-level voltage to a third terminal, and a fourth switch to output a low-level voltage to a fourth terminal, the conductor plates include, in a state of being installed in the connector opening, a first joint member that electrically connects the first terminal and the second terminal, a second joint member that electrically connects the third terminal and the fourth terminal, and the insulating member that physically couples the first joint member and the second joint member in a state in which the first joint member and the second joint member are electrically insulated, the first joint member has a first through hole and a second through hole through which the first terminal and the second terminal are inserted, respectively, the

second joint member has a third through hole and a fourth through hole through which the third terminal and the fourth terminal are inserted, respectively, the first joint member has a first concave portion or a first convex portion that are fitted to the insulating member in a direction intersecting an insertion direction of the terminals, and the second joint member has a second concave portion or a second convex portion that are fitted to the insulating member in the direction intersecting with the insertion direction of the terminals.